

# Microsoft questions

1

## Task 2

C++

Files

2

The effects of global warming are becoming more and more noticeable each year. That is why many countries aim to reduce greenhouse gas emissions by generating electricity from the wind instead of from fossil fuels. Wind turbines have fewer negative effects on the environment and emit less carbon dioxide.

A company has got a contract to build a wind farm on an enclosed, rectangular area of land. The goal is to build as many turbines as possible, to maximize the wind farm's power capacity.

It was recently discovered that the land is inhabited by an endangered species of bird. Thus, the company wants to build the wind turbines at a distance greater than  $K$  from every habitat.

The map of the land is given as a rectangular matrix  $A$  of integers, where birds' habitats are represented by 1 and all other locations are represented by 0.

Once constructed, a turbine occupies exactly one cell of the map and each cell can contain at most one turbine. The distance between two locations is measured as the sum of horizontal and vertical distances between two cells on the map. For example, the distance between cells  $A[3][2]$  and  $A[1][4]$  is  $|1 - 3| + |4 - 2| = 4$ .

Write a function:

task2

solution

test-input

Test Output



Task 2

C++

Files

task2

solution.cp

test-input.t

Test Output

1 The map of the land is given as a rectangular matrix A of integers, where birds' habitats are represented by 1 and all other locations are represented by 0.

2 Once constructed, a turbine occupies exactly one cell of the map and each cell can contain at most one turbine. The distance between two locations is measured as the sum of horizontal and vertical distances between two cells on the map. For example, the distance between cells A[3][2] and A[1][4] is  $|1 - 3| + |4 - 2| = 4$ .

Write a function:

```
int solution(vector<vector<int>> &A, int K);
```

that, given matrix A consisting of N rows and M columns and an integer K, returns the maximum number of wind turbines that can be built while following the environmentalists' advice.

Examples:

1. Given A of size  $4 \times 3$  and  $K = 1$  with bird habitats at (0, 0), (2, 2) and (3, 1):

|         |     |         |     |         |     |
|---------|-----|---------|-----|---------|-----|
| A[0][0] | = 1 | A[0][1] | = 0 | A[0][2] | = 0 |
| A[1][0] | = 0 | A[1][1] | = 0 | A[1][2] | = 0 |
| A[2][0] | = 0 | A[2][1] | = 0 | A[2][2] | = 1 |
| A[3][0] | = 0 | A[3][1] | = 1 | A[3][2] | = 0 |

Task 2

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Files

task2

solution

test-input

Test Output

1. Given A of size  $4 \times 3$  and  $K = 1$  with bird habitats at (0, 0), (2, 2) and (3, 1):

|         |     |         |     |         |     |
|---------|-----|---------|-----|---------|-----|
| A[0][0] | = 1 | A[0][1] | = 0 | A[0][2] | = 0 |
| A[1][0] | = 0 | A[1][1] | = 0 | A[1][2] | = 0 |
| A[2][0] | = 0 | A[2][1] | = 0 | A[2][2] | = 1 |
| A[3][0] | = 0 | A[3][1] | = 1 | A[3][2] | = 0 |

the function should return 3. Wind turbines can be built in three locations: (0, 2), (1, 1) and (2, 0).

|   |                                                                                     |                                                                                     |                                                                                     |
|---|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|   | 0                                                                                   | 1                                                                                   | 2                                                                                   |
| 0 |  |                                                                                     |                                                                                     |
| 1 |                                                                                     |                                                                                     |                                                                                     |
| 2 |                                                                                     |                                                                                     |  |
| 3 |                                                                                     |  |                                                                                     |

2. Given A of size  $3 \times 4$  and  $K = 2$  with no habitats:



There are  $N$  houses along the street. Carbon filters are already installed in some of them. We would like to install filters in the remaining houses (those that do not possess them yet). Two types of filter, named 'a' and 'b', are being used. The filters work best if no three adjacent houses have the same type of filter. The houses are represented as a string of characters 'a', 'b' and '?' ('a' and 'b' denote a house with a filter of a given type installed; '?' represents a house with no filter yet). Your task is to make a plan of the filter types to be installed in the houses that do not yet have them.

Write a function

```
string solution(string &S);
```

that, given a string  $S$  of length  $N$ , returns a string that is the result of replacing each '?' in string  $S$  with an 'a' or a 'b' character and does not contain three identical consecutive letters (in other words, neither "aaa" nor "bbb" may occur in the processed string).

Examples:

1. Given  $S = "a?bb"$ , your function should return "aabb".
2. Given  $S = "??abb"$ , your function should return "ababb", "bbabb" or "baabb".

that, given a string  $S$  of length  $N$ , returns a string that is the result of replacing each '?' in string  $S$  with an 'a' or a 'b' character and does not contain three identical consecutive letters (in other words, neither "aaa" nor "bbb" may occur in the processed string).

Examples:

1. Given  $S = "a?bb"$ , your function should return "aabb".
2. Given  $S = "??abb"$ , your function should return "ababb", "bbabb" or "baabb".
3. Given  $S = "a?b?aa"$ , your function should return "aabbaa".
4. Given  $S = "aa??aa"$ , your function should return "aabbaa".

Write an **efficient** algorithm for the following assumptions:

- string  $S$  consists only of the following characters: "a", "b" and/or "?";
- $N$  is an integer within the range  $[1..500,000]$ ;
- it is always possible to create a plan so that there are no three identical consecutive filters.

Your task is to find the number of vertices on the longest path in the tree, such that no pair of adjacent vertices on the path would contain the same letter.

Write a function:

```
int solution(string &S, vector<int> &A);
```

that, given string S and array A of N integers, returns an integer representing the number of vertices on the longest path in which no two consecutive vertices contain the same letter.

Examples:

1. Given S = "ab" and A = [-1, 0], your function should return 2. The longest path is: 0 - 1. It is marked in red on the picture.

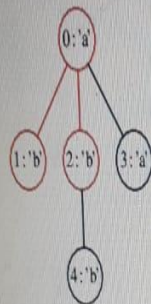


2. Given S = "abbab" and A = [-1, 0, 0, 0, 2], your function should return 3. The longest path is: 1 - 0 - 2. It is marked in red on the

Task 2

C++

2. Given S = "abbab" and A = [-1, 0, 0, 0, 2], your function should return 3. The longest path is: 1 - 0 - 2. It is marked in red on the picture.



3. Given S = "abab" and A = [-1, 2, 0, 1], your function should return 2. The longest path is: 2 - 1. It is marked in red on the picture.



Task 2

C++

3. Given S = "abab" and A = [-1, 2, 0, 1], your function should return 2. The longest path is: 2 - 1. It is marked in red on the picture.



Write an efficient algorithm for the following assumptions:

- N is an integer within the range [1..40,000];
- string S consists only of the characters "a" and/or "b";
- string S and array A are both of length N;
- array A defines a proper tree with a root in node 0.

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1

## Task 1

C++

Files

2

A car manufacturer has data about the production processes of  $N$  different cars (numbered from 0 to  $N-1$ ) and wants to maximize the number of cars produced in the upcoming month. The manufacturing information is described by an array  $H$ , where  $H[K]$  denotes the number of hours required to produce the  $K$ -th car.

There are two assembly lines in the factory, the first of which works for  $X$ , and the second  $Y$ , hours in a month. Every car can be constructed using either one of these lines. Only one car at a time can be produced on each assembly line and it is not possible to switch lines after starting the car's production.

What is the maximum number of different cars that can be produced in the upcoming month?

Write a function:

```
int solution(vector<int> &H, int X, int Y);
```

that, given an array  $H$  of  $N$  integers and two integers  $X$  and  $Y$ , returns the maximum number of different cars that can be produced in the upcoming month by assigning cars to assembly lines in an optimal way.

**Examples:**

1. Given  $H = [1, 1, 3]$ ,  $X = 1$  and  $Y = 1$ , the answer should be 2. Only cars whose assembly time requires 1 hour can be constructed.
2. Given  $H = [6, 5, 5, 4, 3]$ ,  $X = 8$ ,  $Y = 9$ , the answer should be 4. The cars that need 3 and 5 hours can be produced on the first assembly line while the second car that needs 5 hours and the car that needs 4 hours can be produced using the second line.
3. Given  $H = [6, 5, 2, 1, 8]$ ,  $X = 17$ ,  $Y = 5$ , the answer should be 5. The car that needs 5 hours can be produced on the second line and the four other cars can be produced on the first line.
4. Given  $H = [5, 5, 4, 6]$ ,  $X = 8$ ,  $Y = 8$ , the answer should be 2. Only one car can be produced on each line.

Notice that the examples above can have many possible assignments of cars to the assembly lines.

Write an **efficient** algorithm for the following assumptions:

- $N$  is an integer within the range  $[1..1,000]$ ;
- each element of array  $H$  is an integer within the range  $[1..1,000]$ ;

## Task 2

C++

We have a map of an area where two square solar power plants are planned to be built. The map is given as a rectangular matrix A of boolean values.  $A[R][C]$  is true if a place in the R-th row and C-th column can be covered by a solar power plant and false if it is not possible.

Each solar power plant must be built in a square shape. Note that a single cell is sufficient for a valid solar power plant. Solar power plants cannot share any cell and should be of the same size.

What is the area of each of the two biggest solar power plants that can be built?

Write a function:

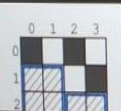
```
int solution(vector< vector<bool> > &A);
```

that, given a matrix A consisting of N rows and M columns, returns the area of the largest possible solar power plants that can be built in a square shape. If no two square solar power plants can be built, the function should return 0.

Examples:

1. Given the  $4 \times 4$  map shown in the figure below, the function should return 4. Cells on which the solar power plant can be built are coloured white and cells that are unavailable are coloured black. Two solar power plants are hatched. The area of each power plant is equal to 4, as they each occupy four cells.

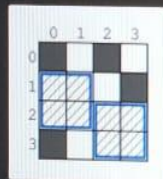
```
A[0][0] = `false`  A[0][1] = `true`  A[0][2] = `false`  A[0][3] = `true`
A[1][0] = `true`   A[1][1] = `true`  A[1][2] = `true`  A[1][3] = `false`
A[2][0] = `true`   A[2][1] = `true`  A[2][2] = `true`  A[2][3] = `true`
A[3][0] = `false`  A[3][1] = `true`  A[3][2] = `true`  A[3][3] = `true`
```



## Task 2

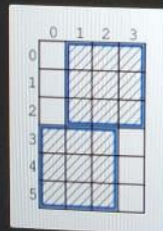
C++

```
A[1][0] = `true`  A[1][1] = `true`  A[1][2] = `true`  A[1][3] = `false`
A[2][0] = `true`  A[2][1] = `true`  A[2][2] = `true`  A[2][3] = `true`
A[3][0] = `false` A[3][1] = `true`  A[3][2] = `true`  A[3][3] = `true`
```



2. Given the  $6 \times 4$  map shown in the figure below, the function should return 9.

```
A[0][0] = `true`  A[0][1] = `true`  A[0][2] = `true`  A[0][3] = `true`
A[1][0] = `true`  A[1][1] = `true`  A[1][2] = `true`  A[1][3] = `true`
A[2][0] = `true`  A[2][1] = `true`  A[2][2] = `true`  A[2][3] = `true`
A[3][0] = `true`  A[3][1] = `true`  A[3][2] = `true`  A[3][3] = `true`
A[4][0] = `true`  A[4][1] = `true`  A[4][2] = `true`  A[4][3] = `true`
A[5][0] = `true`  A[5][1] = `true`  A[5][2] = `true`  A[5][3] = `true`
```



3. Given the  $3 \times 5$  map shown in the figure below, the function should return 0. It is impossible to build two solar power plants.



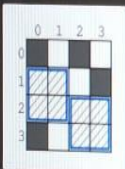
1h 27min

## Task 2

C++

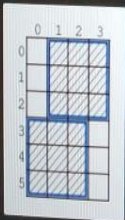
2

```
A[1][0] = 'true'  A[1][1] = 'true'  A[1][2] = 'true'  A[1][3] = 'false'
A[2][0] = 'true'  A[2][1] = 'true'  A[2][2] = 'true'  A[2][3] = 'true'
A[3][0] = 'false' A[3][1] = 'true'  A[3][2] = 'true'  A[3][3] = 'true'
```



2. Given the 6 x 4 map shown in the figure below, the function should return 9.

```
A[0][0] = 'true'  A[0][1] = 'true'  A[0][2] = 'true'  A[0][3] = 'true'
A[1][0] = 'true'  A[1][1] = 'true'  A[1][2] = 'true'  A[1][3] = 'true'
A[2][0] = 'true'  A[2][1] = 'true'  A[2][2] = 'true'  A[2][3] = 'true'
A[3][0] = 'true'  A[3][1] = 'true'  A[3][2] = 'true'  A[3][3] = 'true'
A[4][0] = 'true'  A[4][1] = 'true'  A[4][2] = 'true'  A[4][3] = 'true'
A[5][0] = 'true'  A[5][1] = 'true'  A[5][2] = 'true'  A[5][3] = 'true'
```



3. Given the 3 x 5 map shown in the figure below, the function should return 0. It is impossible to build two solar power plants.

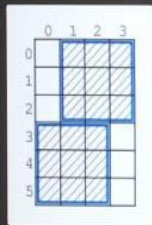
1h 27min

## Task 2

C++

2

```
A[5][0] = 'true'  A[5][1] = 'true'  A[5][2] = 'true'  A[5][3] = 'true'
```



3. Given the 3 x 5 map shown in the figure below, the function should return 0. It is impossible to build two solar power plants.

```
A[0][0] = 'false'  A[0][1] = 'false'  A[0][2] = 'false'  A[0][3] = 'false'  A[0][4] = 'false'
A[1][0] = 'false'  A[1][1] = 'false'  A[1][2] = 'true'  A[1][3] = 'false'  A[1][4] = 'false'
A[2][0] = 'false'  A[2][1] = 'false'  A[2][2] = 'false'  A[2][3] = 'false'  A[2][4] = 'false'
```



Write an efficient algorithm for the following assumptions:

- N and M are integers within the range [1..700].



1h 22min

## Task 2

C++

Files

task2

solution.cpp

test-input.txt

We have a map of an area where two square solar power plants are planned to be built. The map is given as a rectangular matrix  $A$  of boolean values.  $A[R][C]$  is `true` if a place in the  $R$ -th row and  $C$ -th column can be covered by a solar power plant and `false` if it is not possible.

Each solar power plant must be built in a square shape. Note that a single cell is sufficient for a valid solar power plant. Solar power plants cannot share any cell and should be of the same size.

What is the area of each of the two biggest solar power plants that can be built?

Write a function:

```
int solution(vector<vector<bool>> &A);
```

that, given a matrix  $A$  consisting of  $N$  rows and  $M$  columns, returns the area of the largest possible solar power plants that can be built in a square shape. If no two square solar power plants can be built, the function should return 0.

Examples:

- Given the  $4 \times 4$  map shown in the figure below, the function should return 4.
- Cells on which the solar power plant can be built are coloured white and cells that are unavailable are coloured black. Two solar power plants are hatched. The area of each power plant is equal to 4, as they each occupy four cells.

Test Output

1h 22min

## Task 2

C++

Files

task2

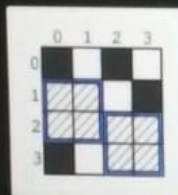
solution.cpp

test-input.txt

examples:

- Given the  $4 \times 4$  map shown in the figure below, the function should return 4.
- Cells on which the solar power plant can be built are coloured white and cells that are unavailable are coloured black. Two solar power plants are hatched. The area of each power plant is equal to 4, as they each occupy four cells.

```
A[0][0] = `false`  A[0][1] = `true`  A[0][2] = `false`  A[0][3] = `true`
A[1][0] = `true`   A[1][1] = `true`  A[1][2] = `true`  A[1][3] = `false`
A[2][0] = `true`   A[2][1] = `true`  A[2][2] = `true`  A[2][3] = `true`
A[3][0] = `false`  A[3][1] = `true`  A[3][2] = `true`  A[3][3] = `true`
```



- Given the  $6 \times 4$  map shown in the figure below, the function should return 9.

```
A[0][0] = `true`  A[0][1] = `true`  A[0][2] =
```

Test Output

1

Task 1

C++

Files

2

There are  $N$  houses along the street. Carbon filters are already installed in some of them. We would like to install filters in the remaining houses (those that do not possess them yet). Two types of filter, named 'a' and 'b', are being used. The filters work best if no three adjacent houses have the same type of filter. The houses are represented as a string of characters 'a', 'b' and '?' ('a' and 'b' denote a house with a filter of a given type installed; '?' represents a house with no filter yet). Your task is to make a plan of the filter types to be installed in the houses that do not yet have them.

Write a function

```
string solution(string &S);
```

that, given a string  $S$  of length  $N$ , returns a string that is the result of replacing each '?' in string  $S$  with an 'a' or a 'b' character and does not contain three identical consecutive letters (in other words, neither "aaa" nor "bbb" may occur in the processed string).

Examples:

1. Given  $S = "a?bb"$ , your function should return "aabb".
2. Given  $S = "??abb"$ , your function should return "ababb", "bbabb" or "baabb".
3. Given  $S = "a?b?aa"$ , your function should return "aabbbaa".

task1

so

tes

Test Output



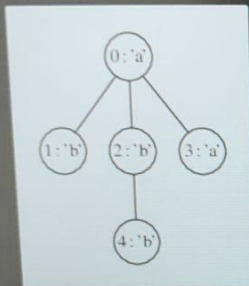
## Task 2

C++

You are given a tree consisting of  $N$  nodes, numbered from 0 to  $N-1$ . Each node contains one of the letters 'a' or 'b'.

The tree is represented as an array  $A$  of length  $N$ .  $A[K]$  (for  $K$  from 0 to  $N-1$ ) denotes the parent of the  $K$ -th node. Node 0 is the root node and does not have a parent, so the value corresponding to it in array  $A$  will always be -1. Letters in nodes are represented as a string  $S$ .  $S[K]$  (for  $K$  from 0 to  $N-1$ ) denotes the letter in the  $K$ -th node.

For example,  $S = \text{"abbab"}$  and  $A = [-1, 0, 0, 0, 2]$  represents a tree made of five vertices. It contains following edges: 0 - 1, 0 - 2, 0 - 3, 2 - 4. The picture below illustrates the tree.



Your task is to find the number of vertices on the longest path in the tree, such that no pair of adjacent vertices on the path would contain the same letter.

Write a function:

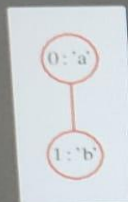
## Task 2

C++

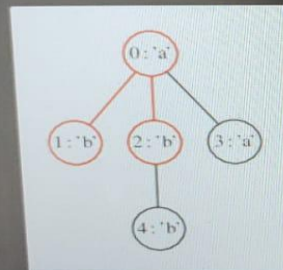
vertices contain the same letter.

Examples:

1. Given  $S = \text{"ab"}$  and  $A = [-1, 0]$ , your function should return 2. The longest path is: 0 - 1. It is marked in red on the picture.



2. Given  $S = \text{"abbab"}$  and  $A = [-1, 0, 0, 0, 2]$ , your function should return 3. The longest path is: 1 - 0 - 2. It is marked in red on the picture.



3. Given  $S = \text{"abab"}$  and  $A = [-1, 2, 0, 1]$ , your function should return 2. The longest path is: 2 - 1. It is marked in red on the picture.

0h 56min

## 1 Task 1

Java 8

Files

solution

2

A car manufacturer has data about the production processes of  $N$  different cars (numbered from 0 to  $N-1$ ) and wants to maximize the number of cars produced in the upcoming month. The manufacturing information is described by an array  $H$ , where  $H[K]$  denotes the number of hours required to produce the  $K$ -th car.

There are two assembly lines in the factory, the first of which works for  $X$ , and the second  $Y$ , hours in a month. Every car can be constructed using **either one** of these lines. Only one car at a time can be produced on each assembly line and it is not possible to switch lines after starting the car's production.

What is the maximum number of **different** cars that can be produced in the upcoming month?

Write a function:

```
class Solution { public int solution(int[] H, int X, int Y); }
```

that, given an array  $H$  of  $N$  integers and two integers  $X$  and  $Y$ , returns the maximum number of different cars that can be produced in the upcoming month by assigning cars to assembly lines in an optimal way.



Examples:



1. Given  $H = [1, 1, 3]$ ,  $X = 1$  and  $Y = 1$ , the answer should be 2. Only cars whose assembly time requires 1 hour can be constructed.



2. Given  $H = [6, 5, 5, 4, 3]$ ,  $X = 8$ ,  $Y = 9$ , the answer should be 4. The cars that need 3 and 5 hours can be

Test Output

|         |    |     |
|---------|----|-----|
| task1   | 1  | //  |
| sol...  | 2  | //  |
| test... | 3  |     |
|         | 4  | //  |
|         | 5  | //  |
|         | 6  |     |
|         | 7  | cla |
|         | 8  |     |
|         | 9  | int |
|         | 10 | Jav |
|         | 11 | }   |
|         | 12 |     |



1h 31min

Submit Task

## Task 2

Java 8

Files

solution.java x

Our task concerns a city with  $N$  roads connecting  $N + 1$  junctions numbered from 0 to  $N$ . Roads connect junctions in such a way that each distinct pair of junctions is connected either by a direct road or through a path consisting of direct roads. There is exactly one way to reach any junction from any other junction.

There is an office next to junction 0 and a house next to every other junction. One person lives in each house. All the citizens work in the office and use a car to travel from their houses to the office. Each car has five *seats* and consumes 1 liter of fuel to move by one road (from one junction to a neighboring junction).

The citizens want to set up a carpool scheme that allows them all to travel to the office in a way that minimizes the total consumption of fuel. People can change cars at each house and then continue to travel together. Calculate the fuel consumption in the optimal scheme.

The network of cities is described using arrays  $A$  and  $B$ , each of length  $N$ . Each pair  $A[K]$ ,  $B[K]$  for  $0 \leq K < N$  specifies a road connecting junctions number  $A[K]$  and  $B[K]$ .

Write a function:

```
class Solution { public int solution(int[] A, int[] B); }
```

that, given non-empty arrays  $A$  and  $B$  of length  $N$  describing a road network in the city, returns the minimum fuel consumption for the optimal carpool scheme.

Examples:

1. Given  $A = [0, 1, 1]$  and  $B = [1, 2, 2]$  there are  $N = 3$  houses, as in the image below. The function should return 2, as

task2

solu...

test...

```
1 // you can also use imports, for
  example:
2 // import java.util.*;
3
4 // you can write to stdout for
  debugging purposes, e.g.
5 // System.out.println("this is a debug
  message");
6
7 class Solution {
8     public int solution(int[] A, int[]
  B) {
9         // write your code in Java SE 8
10    }
11 }
12
```

Test Output

Run Tests

## Task 2

C++

Our task concerns a city with  $N$  roads connecting  $N + 1$  junctions numbered from 0 to  $N$ . Roads connect junctions in such a way that each distinct pair of junctions is connected either by a direct road or through a path consisting of direct roads. There is exactly one way to reach any junction from any other junction.

There is an office next to junction 0 and a house next to every other junction. One person lives in each house. All the citizens work in the office and use a car to travel from their houses to the office. Each car has five *seats* and consumes one liter of fuel to move by one road (from one junction to a neighboring junction).

The citizens want to set up a carpool scheme that allows them all to travel to the office in a way that minimizes the total consumption of fuel. People can change cars at each house and then continue to travel together. Calculate the fuel consumption in the optimal scheme.

The network of cities is described using arrays  $A$  and  $B$ , each of length  $N$ . Each pair  $A[K], B[K]$  for  $0 \leq K < N$  specifies a road connecting junctions number  $A[K]$  and  $B[K]$ .

Write a function:

```
int solution(vector<int> &A, vector<int> &B);
```



1

Task 1

Java 8

2

You are given a positive integer  $N$ . Your task is to find the smallest integer greater than  $N$  that does not contain two identical consecutive digits.

For example, given  $N = 1765$ , the smallest integer greater than  $N$  is 1766. However, in 1766 the last two digits are identical. The next integer, 1767, does not contain two identical consecutive digits, and is the smallest integer greater than 1765 that fulfils the condition. Note that the second and fourth digits in 1767 can both be 7 as they are not consecutive.

Write a function:

```
class Solution { public int solution(int N); }
```

that, given an integer  $N$ , returns the smallest integer greater than  $N$  that does not contain two identical consecutive digits.

Examples:

1. Given  $N = 55$ , the function should return 56. It is the smallest integer greater than 55 and it does not contain two consecutive digits that are the same.
2. Given  $N = 1765$ , the function should return 1767, as explained

1

Task 1

Java 8

2

```
class Solution { public int solution(int N); }
```

that, given an integer  $N$ , returns the smallest integer greater than  $N$  that does not contain two identical consecutive digits.

Examples:

1. Given  $N = 55$ , the function should return 56. It is the smallest integer greater than 55 and it does not contain two consecutive digits that are the same.
2. Given  $N = 1765$ , the function should return 1767, as explained above.
3. Given  $N = 98$ , the answer should be 101. Both 99 and 100 contain two identical consecutive digits, but 101 does not.
4. Given  $N = 44432$ , the answer should be 45010.
5. Given  $N = 3298$ , the answer should be 3401.

Write an efficient algorithm for the following assumptions:

- $N$  is an integer within the range  $[1..1,000,000,000]$ .

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All changes saved

## Task 2

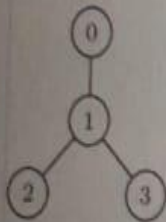
```
class Solution { public int solution(int[] A, int[] B); }
```

that, given non-empty arrays A and B of length N describing a road network in the city, returns the minimum fuel consumption for the optimal carpool scheme.

## Examples:

1. Given A = [0, 1, 1] and B = [1, 2, 3], there are N = 3 houses, as in the image below. The function should return 3, as the optimal carpool scheme uses 3 liters of fuel:

- Citizen 2 goes in his car from his house to house 1.
- Citizen 3 goes in her car from her house to house 1.
- Citizens 2 and 3 go with citizen 1 in his car to the office.



2. Given A = [1, 1, 1, 9, 9, 9, 9, 7, 8] and B = [2, 0, 3, 1, 6, 5, 4, 0, 0], there are N = 9 houses, as in the image below. The function should return 10, as the following carpool scheme is optimal:

- Citizens 7 and 8 go directly to the office in their cars (using 2 liters of fuel).
- Citizens 4, 5 and 6 go in their cars to house 0 (they use 2 liters of fuel) and then they go with citizen 0

Java B

Files

solution.java x

task2

solu...

test...

```

1 // you can also use imports, for
  example:
2 // import java.util.*;
3
4 // you can write to stdout for
  debugging purposes, e.g.
5 // System.out.println("this is a
  message");
6
7 class Solution {
8     public int solution(int[] A,
9     B) {
10         // write your code in Java
11     }
12

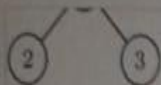
```

Test Output

Run

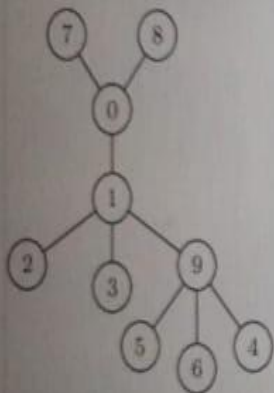


2



2. Given  $A = [1, 1, 1, 9, 9, 9, 7, 8]$  and  $B = [2, 0, 3, 1, 6, 5, 4, 0, 0]$ , there are  $N = 9$  houses, as in the image below. The function should return 10, as the following carpool scheme is optimal:

- Citizens 7 and 8 go directly to the office in their cars (using 2 liters of fuel).
- Citizens 4, 5, and 6 go in their cars to house 9 (they use 3 liters of fuel) and then they go with citizen 9 in her car to house 1 (for another 1 liter).
- Citizens 2 and 3 go in their cars to house 1 (for 2 liters of fuel).
- Now 7 citizens are at house 1, so they need to use at least two cars to move to the office (for another 2 liters of fuel).



Write an efficient algorithm for the following question:

Java 8

Files

solution.java x

task2

solu...

test...

```

1 // you can also use imports, for
  // example:
2 // import java.util.*;
3
4 // you can write to stdout for
  // debugging purposes, e.g.
5 // System.out.println("this is a
  // message");
6
7 class Solution {
8     public int solution(int[] A,
9         B) {
10         // write your code in Java
11         here
12     }
  
```

Test Output

Run